

MAHARISHI MARKANDESHWAR (DEEMED TO BE UNIVERSITY)

MM ENGINEERING COLLEGE (MMEC)

Mullana, Ambala, Haryana - 133207 • Accredited by NAAC with Grade 'A++'



OFFICIAL LABORATORY RECORD & MANUAL

Fundamental of Web Technologies Practical Manual

Course Code: BCSE-013 • Semester 3 / 4 (2nd Year)

Department of Computer Science & Engineering • Session 2025-26

STUDENT PRACTICAL RECORD & EVALUATION DOSSIER

Student Name:

Roll Number:

University Reg No:

Class & Section:

Academic Program: CSE / IT / AI-DS / AI-ML

Curriculum Batch: 2025 - 2029 (Session 2025-26)

Signature of Laboratory Faculty Incharge

Signature of Head of Department

Exp #	Title / Aim of the Practical Experiment	Date of Perf.	Date of Subm.	Marks	Sign
1	Exploration of Web Architecture, Browsers, Web Servers & WordPress Installation	/..../2025	/..../2025	/ 10	
2	Biography Webpage of Idol (Dr. APJ Abdul Kalam) with Headings & Horizontal Rules	/..../2025	/..../2025	/ 10	
3	Ordered & Unordered Nested Lists with Multi-Level Formatting	/..../2025	/..../2025	/ 10	
4	Interactive Country Image Map with Clickable Hotspot Coordinates	/..../2025	/..../2025	/ 10	
5	Academic Resume / Bio-Data using Semantic HTML Table with Rowspan & Colspan	/..../2025	/..../2025	/ 10	
6	Three-Zone Screen Layout (15% Top Banner, 70% Content, 15% Lower Menu)	/..../2025	/..../2025	/ 10	
7	Student Registration Form with HTML5 & JavaScript Regex Validation	/..../2025	/..../2025	/ 10	
8	Multimedia Tag Integration (<video> & <audio> with Fallback Sources)	/..../2025	/..../2025	/ 10	
9	CSS3 Box Model, <div> Containers, Box-Shadow & Interactive Hover Transitions	/..../2025	/..../2025	/ 10	
10	Modern CSS3 Styling (Gradients, Link Pseudo-States, Image Opacity & Borders)	/..../2025	/..../2025	/ 10	

Exploration of Web Architecture, Browsers, Web Servers & WordPress Installation

AIM / OBJECTIVE:

Conduct in-depth architectural study of the World Wide Web: Client-Server HTTP/HTTPS handshake, Web Browsers vs Web Servers (Apache/Nginx), and local installation/setup of WordPress CMS on XAMPP/WAMP.

ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Study 3-tier web architecture: Presentation (Client), Application (Web Server), Data (Database).
- Step 2: Understand HTTP Request-Response cycle, status codes (200, 301, 404, 500).
- Step 3: Setup local development environment using XAMPP/WAMP (Apache + MySQL).
- Step 4: Download WordPress ZIP, place in htdocs, create MySQL database wp_db.
- Step 5: Complete 5-minute WordPress install wizard and document dashboard components.

SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<?php
// wp-config.php - WordPress Core Database Configuration
define( 'DB_NAME', 'mmecc_wordpress_db' );
define( 'DB_USER', 'root' );
define( 'DB_PASSWORD', '' );
define( 'DB_HOST', 'localhost' );
define( 'DB_CHARSET', 'utf8mb4' );
define( 'DB_COLLATE', '' );

define( 'AUTH_KEY',          'put-your-unique-phrase-here-1234567890' );
define( 'SECURE_AUTH_KEY',   'put-your-unique-phrase-here-0987654321' );

$table_prefix = 'wp_';
define( 'WP_DEBUG', false );

if ( ! defined( 'ABSPATH' ) ) {
    define( 'ABSPATH', __DIR__ . '/' );
}
require_once ABSPATH . 'wp-settings.php';
?>
```

SAMPLE INPUT:

Localhost URL: <http://localhost/wordpress>

EXPECTED OUTPUT:

WordPress 6.x CMS installed with local Apache/MySQL running.

Biography Webpage of Idol (Dr. APJ Abdul Kalam) with Headings & Horizontal Rules

🎯 AIM / OBJECTIVE:

Design a clean, semantic HTML5 biography webpage of an idol (Dr. APJ Abdul Kalam) utilizing all heading tags (<h1> through <h6>), paragraphs, blockquotes, and horizontal dividing rules (<hr>).

⚙️ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Set up HTML5 doctype and meta viewport.
- Step 2: Use <h1> for main idol name.
- Step 3: Use <h2> and <h3> for Early Life, Scientific Career, Presidency, and Awards.
- Step 4: Insert <hr> dividers between major life chapters.
- Step 5: Embed inspirational quote using <blockquote>.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Biography - Dr. A.P.J. Abdul Kalam (Missile Man of India)</title>
  <style>
    body { font-family: 'Georgia', serif; line-height: 1.8; margin: 40px auto; max-width: 800px; color: #2d3748; background-color: #f7fafc; padding: 20px; }
    h1 { color: #1a365d; text-align: center; font-size: 2.2rem; }
    h2 { color: #2b6cb0; border-bottom: 2px solid #cbd5e0; padding-bottom: 6px; }
    h3 { color: #2c5282; }
    hr { border: 0; height: 1px; background-image: linear-gradient(to right, rgba(0,0,0,0), rgba(0,0,0,0.3), rgba(0,0,0,0)); margin: 25px 0; }
    blockquote { font-style: italic; background: #edf2f7; border-left: 5px solid #3182ce; padding: 12px 20px; margin: 20px 0; }
  </style>
</head>
<body>
  <h1>Dr. Avul Pakir Jainulabdeen Abdul Kalam</h1>
  <p style="text-align:center; font-weight:bold; color:#4a5568;">15 October 1931 - 27 July 2015 | 11th President of India</p>
  <hr>

  <h2>1. Early Life & Humble Beginnings</h2>
  <p>Dr. Kalam was born in Rameswaram, Tamil Nadu. Coming from a poor background, he distributed newspapers in his early childhood to support his family's income while pursuing his deep passion for mathematics and aeronautical engineering.</p>

  <hr>
  <h2>2. Career as an Aerospace Scientist</h2>
  <h3>2.1 Indian Space Research Organisation (ISRO)</h3>
  <p>He served as the Project Director of India's first satellite launch vehicle (SLV-III) which successfully deployed the Rohini satellite into near-Earth orbit in 1980.</p>

  <h3>2.2 Defence Research and Development Organisation (DRDO)</h3>
  <p>He led the Integrated Guided Missile Development Programme (IGMDP), spearheading missiles like AGNI and PRITHVI, earning the title <strong>"The Missile Man of India"</strong>.</p>

  <hr>
  <h2>3. Presidency (2002 - 2007)</h2>
  <p>Affectionately known as the <em>"People's President"</em>, he transformed Rashtrapati Bhavan into a hub for students, innovators, and thinkers across the nation.</p>

  <blockquote>
    "Dream, dream, dream. Dreams transform into thoughts and thoughts result in action."
    <br>&mdash; <strong>Dr. A.P.J. Abdul Kalam</strong>
  </blockquote>
```

```
</body>  
</html>
```

**SAMPLE INPUT:**

Browser loading biography.html

**EXPECTED OUTPUT:**

Rendered elegant biography with h1-h6 hierarchy and horizontal rules.

Ordered & Unordered Nested Lists with Multi-Level Formatting

🎯 AIM / OBJECTIVE:

Create an HTML webpage demonstrating nested ordered lists (), unordered lists (), and definition lists (<dl>) with customized bullet types (disc, circle, square, roman numerals, letters).

⚙️ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Open with type='I' for Roman numerals.
- Step 2: Inside , nest unordered list with type='square'.
- Step 3: Inside nested item, nest ordered list with type='a' for alphabets.
- Step 4: Include a definition list (<dl>, <dt>, <dd>) for technical glossary.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 3 - Nested Lists in HTML</title>
  <style>
    body { font-family: Arial, sans-serif; padding: 25px; line-height: 1.6; }
    h2 { color: #1e3a8a; }
    ol { margin-bottom: 15px; }
    dt { font-weight: bold; color: #0d9488; }
    dd { margin-left: 20px; margin-bottom: 10px; }
  </style>
</head>
<body>
  <h2>Computer Science Engineering - Curriculum Roadmap</h2>
  <ol type="I">
    <li><strong>Year 1: Foundation Courses</strong>
      <ul style="list-style-type: square;">
        <li>Programming in C (BCSE-008 / BCSE-001)
          <ol type="a">
            <li>Control Structures & Loops</li>
            <li>Pointers & Dynamic Memory</li>
            <li>File Handling</li>
          </ol>
        </li>
        <li>Basics of Electrical & Electronics (BELE-001)</li>
      </ul>
    </li>
    <li><strong>Year 2: Core Engineering</strong>
      <ul style="list-style-type: disc;">
        <li>Data Structures & Algorithms (BCSE-007)</li>
        <li>Web Technologies (BCSE-012 / BCSE-013)</li>
      </ul>
    </li>
  </ol>

  <h2>Web Tech Glossary (Definition List)</h2>
  <dl>
    <dt>HTML5</dt>
    <dd>HyperText Markup Language - Standard markup language for documents designed to be displayed in a web browser.</dd>
    <dt>CSS3</dt>
    <dd>Cascading Style Sheets - Language used for describing the presentation and styling of a document.</dd>
  </dl>
</body>
</html>
```

🖥️ SAMPLE INPUT:

[View in Chrome / Firefox](#)

🖥️ EXPECTED OUTPUT:

Clean multi-level nested list hierarchy with varied bullet & numbering types.

Interactive Country Image Map with Clickable Hotspot Coordinates

🎯 AIM / OBJECTIVE:

Develop an interactive image map webpage using with usemap attribute and <map> with <area> tags (rect, circle, poly) linking specific geographic regions to information pages.

⚙ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Embed map image with usemap='#worldmap'.
- Step 2: Define <map name='worldmap'>.
- Step 3: Define <area shape='rect' coords='x1,y1,x2,y2' href='...' alt='Region'>.
- Step 4: Define <area shape='circle' coords='x,y,radius' href='...' alt='Capital'>.
- Step 5: Define <area shape='poly' coords='x1,y1,x2,y2,...' href='...' alt='State'>.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 4 - Interactive Image Map</title>
  <style>
    body { font-family: sans-serif; text-align: center; padding: 20px; }
    .map-container { display: inline-block; border: 3px solid #3b82f6; border-radius: 8px; overflow: hidden; }
  </style>
</head>
<body>
  <h2>Interactive Campus & Geography Map</h2>
  <p>Hover and click on specific hotspots (Library, Admin Block, Tech Park) to navigate.</p>

  <div class="map-container">
    

    <map name="campusmap">
      <!-- Rectangular Hotspot for Admin Block -->
      <area shape="rect" coords="50,50,200,180"
        href="#admin"
        alt="Administrative Block"
        onclick="alert('Navigating to: Administrative Block (Registrar & Dean Office)'); return false;">

      <!-- Circular Hotspot for Central Library -->
      <area shape="circle" coords="450,150,75"
        href="#library"
        alt="Central Library"
        onclick="alert('Navigating to: Central Digital Library (50,000+ Volumes)'); return false;">

      <!-- Polygonal Hotspot for Sports Complex -->
      <area shape="poly" coords="250,250,350,220,400,320,300,360,220,310"
        href="#sports"
        alt="Sports Complex"
        onclick="alert('Navigating to: University Olympic Sports Complex'); return false;">
    </map>
  </div>
</body>
</html>
```

🖱 SAMPLE INPUT:

Clicking Library hotspot (coords 450,150)

🖱 EXPECTED OUTPUT:

Click triggers specific region action & coordinates binding.

Academic Resume / Bio-Data using Semantic HTML Table with Rowspan & Colspan

🎯 AIM / OBJECTIVE:

Design a professional academic resume / bio-data table in HTML demonstrating semantic table elements (<thead>, <tbody>, <tfoot>), rowspan, and colspan attributes with alternating row styling.

⚙️ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Create <table> with border, cell-padding, and cell-spacing styling.
- Step 2: Use <caption> for resume title.
- Step 3: Merge columns using colspan for section banners (Personal Info, Education, Skills).
- Step 4: Merge rows using rowspan for multi-semester grade summaries.
- Step 5: Apply clean CSS zebra striping.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 5 - Student Bio-Data Table</title>
  <style>
    body { font-family: 'Segoe UI', Tahoma, sans-serif; padding: 25px; background: #f8fafc; }
    table { width: 100%; max-width: 850px; margin: 0 auto; border-collapse: collapse; background: white; box-shadow:
0 4px 6px rgba(0,0,0,0.08); }
    caption { font-size: 1.5rem; font-weight: bold; margin-bottom: 12px; color: #1e3a8a; }
    th, td { border: 1px solid #cbd5e1; padding: 12px 16px; text-align: left; }
    th { background-color: #2563eb; color: white; }
    .section-header { background-color: #e2e8f0; font-weight: bold; color: #0f172a; text-transform: uppercase; }
    tr:nth-child(even) { background-color: #f1f5f9; }
  </style>
</head>
<body>
  <table>
    <caption>ACADEMIC CURRICULUM VITAE & BIO-DATA</caption>
    <tr>
      <th colspan="4" class="section-header">1. Personal Details</th>
    </tr>
    <tr>
      <td rowspan="2" style="width:140px; text-align:center;">
        
      </td>
      <td><strong>Full Name:</strong> Bhavya Gupta</td>
      <td><strong>Roll No:</strong> 11232541</td>
      <td><strong>Batch:</strong> 2024-2028</td>
    </tr>
    <tr>
      <td><strong>Degree:</strong> B.Tech CSE</td>
      <td colspan="2"><strong>University:</strong> MMEC, Maharishi Markandeshwar University</td>
    </tr>
    <tr>
      <th colspan="4" class="section-header">2. Academic Qualifications</th>
    </tr>
    <tr style="background:#3b82f6; color:white;">
      <td><strong>Exam Passed</strong></td>
      <td><strong>Board / University</strong></td>
      <td><strong>Passing Year</strong></td>
      <td><strong>Percentage / CGPA</strong></td>
    </tr>
    <tr>
      <td>B.Tech (CSE)</td>
      <td>MM(DU) Mullana</td>
      <td>2028 (Pursuing)</td>
      <td>9.24 CGPA</td>
    </tr>
  </table>
```

```

<tr>
  <td>Class XII (CBSE)</td>
  <td>Central Board of Secondary Education</td>
  <td>2024</td>
  <td>91.4%</td>
</tr>
<tr>
  <td>Class X (CBSE)</td>
  <td>Central Board of Secondary Education</td>
  <td>2022</td>
  <td>93.8%</td>
</tr>

<tr>
  <th colspan="4" class="section-header">3. Technical Proficiencies</th>
</tr>
<tr>
  <td><strong>Programming:</strong></td>
  <td colspan="3">C, C++, Python, JavaScript (ES6+), SQL</td>
</tr>
<tr>
  <td><strong>Web Technologies:</strong></td>
  <td colspan="3">HTML5, CSS3, Tailwind CSS, React.js, Node.js</td>
</tr>
</table>
</body>
</html>

```



SAMPLE INPUT:

Browser loading table.html



EXPECTED OUTPUT:

Professional tabular resume with rowspan and colspan bindings.

Three-Zone Screen Layout (15% Top Banner, 70% Content, 15% Lower Menu)

🎯 AIM / OBJECTIVE:

Design a web page layout divided into three horizontal zones: 15% Top Title zone, 70% Middle Content display zone, and 15% Lower Menu navigation zone using modern CSS Flexbox / Grid.

⚙ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Set height: 100vh and margin: 0 on body.
- Step 2: Create .zone-top with height: 15vh (display title & branding).
- Step 3: Create .zone-middle with height: 70vh and overflow-y: auto (scrollable main content).
- Step 4: Create .zone-bottom with height: 15vh (quick action bottom navigation).
- Step 5: Ensure responsive resizing across desktop and mobile screens.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 6 - Three-Zone Page Layout</title>
  <style>
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body { height: 100vh; display: flex; flex-direction: column; font-family: 'Segoe UI', sans-serif; overflow: hidden; }

    .zone-top {
      height: 15vh;
      background: linear-gradient(90deg, #1e40af, #3b82f6);
      color: white;
      display: flex;
      align-items: center;
      justify-content: space-between;
      padding: 0 2rem;
      box-shadow: 0 2px 10px rgba(0,0,0,0.15);
    }

    .zone-middle {
      height: 70vh;
      background: #f8fafc;
      padding: 2rem;
      overflow-y: auto;
    }

    .zone-bottom {
      height: 15vh;
      background: #0f172a;
      color: white;
      display: flex;
      align-items: center;
      justify-content: space-around;
      border-top: 2px solid #334155;
    }

    .nav-item { color: #cbd5e1; text-decoration: none; font-size: 1.1rem; font-weight: 600; padding: 8px 16px; border-radius: 6px; transition: background 0.2s; }
    .nav-item:hover { background: #334155; color: white; }
  </style>
</head>
<body>
  <div class="zone-top">
    <h2>MMEC CSE Digital Portal</h2>
    <span>Academic Year 2025-26</span>
  </div>

  <div class="zone-middle">
```

```
<h3>Zone 2: Main Middle Display Area (70% Height)</h3>
<p style="margin-top: 10px;">This zone holds dynamic course content, lab experiments, video lectures, and syllabus documentation with independent scrollbar.</p>
</div>

<div class="zone-bottom">
  <a href="#home" class="nav-item">🏠 Home</a>
  <a href="#syllabus" class="nav-item">📖 Syllabus</a>
  <a href="#practicals" class="nav-item">🖥️ Practicals</a>
  <a href="#viva" class="nav-item">🎤 Viva Q&A</a>
</div>
</body>
</html>
```

**SAMPLE INPUT:**

Browser viewport 1920x1080

**EXPECTED OUTPUT:**

Exact 15vh header, 70vh scrollable body, and 15vh footer menu.

Student Registration Form with HTML5 & JavaScript Regex Validation

🎯 AIM / OBJECTIVE:

Develop a student registration form collecting Name, Roll No, Email, Phone, Password, and Branch with real-time JavaScript regular expression validation and user feedback.

⚙ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Create HTML <form> with inputs for Name, Email, Password, Phone, Branch.
- Step 2: Attach onsubmit event handler to form.
- Step 3: Validate Name (letters only, min 3 chars).
- Step 4: Validate Email using regex pattern.
- Step 5: Validate Phone (10 digits starting with 6-9).
- Step 6: Validate Password (min 8 chars, 1 number, 1 special character).

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 7 - Validated Registration Form</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; background: #eef2ff; padding: 30px; }
    .form-card { max-width: 480px; margin: 0 auto; background: white; padding: 25px 30px; border-radius: 12px; box-s
hadow: 0 4px 15px rgba(0,0,0,0.08); }
    .form-group { margin-bottom: 16px; }
    label { display: block; margin-bottom: 6px; font-weight: 600; color: #1e293b; }
    input, select { width: 100%; padding: 10px; border: 1px solid #cbd5e1; border-radius: 6px; font-size: 0.95rem; b
ox-sizing: border-box; }
    .error { color: #dc2626; font-size: 0.85rem; margin-top: 4px; display: none; }
    .btn-submit { width: 100%; padding: 12px; background: #2563eb; color: white; border: none; border-radius: 6px; f
ont-size: 1rem; font-weight: bold; cursor: pointer; }
    .btn-submit:hover { background: #1d4ed8; }
  </style>
</head>
<body>
  <div class="form-card">
    <h2 style="text-align:center; margin-bottom:20px; color:#1e40af;">Student Registration</h2>
    <form id="regForm" onsubmit="return validateForm()">
      <div class="form-group">
        <label for="name">Full Name</label>
        <input type="text" id="name" placeholder="Enter your full name">
        <div id="nameErr" class="error">Name must contain only letters (min 3 chars).</div>
      </div>
      <div class="form-group">
        <label for="email">College Email</label>
        <input type="email" id="email" placeholder="student@mmumullana.org">
        <div id="emailErr" class="error">Enter a valid institutional email address.</div>
      </div>
      <div class="form-group">
        <label for="phone">Mobile Number</label>
        <input type="tel" id="phone" placeholder="10-digit mobile number">
        <div id="phoneErr" class="error">Enter valid 10-digit Indian phone (starts 6-9).</div>
      </div>
      <div class="form-group">
        <label for="pass">Password</label>
        <input type="password" id="pass" placeholder="Minimum 8 characters">
        <div id="passErr" class="error">Must be 8+ chars with at least 1 number and 1 special char.</div>
      </div>
      <button type="submit" class="btn-submit">Register Account</button>
    </form>
  </div>

  <script>
```

```

function validateForm() {
    let isValid = true;
    const name = document.getElementById('name').value.trim();
    const email = document.getElementById('email').value.trim();
    const phone = document.getElementById('phone').value.trim();
    const pass = document.getElementById('pass').value;

    if (!/^[a-zA-Z ]{3,40}$/.test(name)) {
        document.getElementById('nameErr').style.display = 'block';
        isValid = false;
    } else document.getElementById('nameErr').style.display = 'none';

    if (!/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/.test(email)) {
        document.getElementById('emailErr').style.display = 'block';
        isValid = false;
    } else document.getElementById('emailErr').style.display = 'none';

    if (!/^[6-9]\d{9}$/.test(phone)) {
        document.getElementById('phoneErr').style.display = 'block';
        isValid = false;
    } else document.getElementById('phoneErr').style.display = 'none';

    if (!/^(?=.*[0-9])(?=.*[!@#%&*])[a-zA-Z0-9!@#%&*]{8,}$/.test(pass)) {
        document.getElementById('passErr').style.display = 'block';
        isValid = false;
    } else document.getElementById('passErr').style.display = 'none';

    if (isValid) {
        alert('🎉 Validation Passed! Student account registered successfully.');
```

```

    }
    return false;
}

```

```

</script>
</body>
</html>

```

SAMPLE INPUT:

Invalid mobile '12345'

EXPECTED OUTPUT:

Instant red validation error message displayed.

Multimedia Tag Integration (<video> & <audio> with Fallback Sources)

🎯 AIM / OBJECTIVE:

Embed native HTML5 audio and video players with multiple codec sources (MP4, WebM, MP3, OGG), custom controls, poster images, and fallback messages.

⚙️ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Use <video controls poster='...'> element.
- Step 2: Add multiple <source> tags (MP4 and WebM) for cross-browser fallback.
- Step 3: Add <audio controls> with MP3 and OGG fallback sources.
- Step 4: Include fallback message for older non-HTML5 compliant browsers.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 8 - HTML5 Multimedia</title>
  <style>
    body { font-family: sans-serif; max-width: 750px; margin: 30px auto; padding: 20px; }
    .media-box { background: #f1f5f9; padding: 20px; border-radius: 10px; margin-bottom: 25px; }
    video, audio { width: 100%; border-radius: 8px; margin-top: 10px; }
  </style>
</head>
<body>
  <h2>HTML5 Multimedia Demonstration</h2>

  <div class="media-box">
    <h3>1. Video Player with Multiple Source Codecs</h3>
    <video controls poster="https://images.unsplash.com/photo-1517694712202-14dd9538aa97?w=600">
      <source src="https://interactive-examples.mdn.mozilla.net/media/cc0-videos/flower.mp4" type="video/mp4">
      <source src="https://interactive-examples.mdn.mozilla.net/media/cc0-videos/flower.webm" type="video/webm">
      Your browser does not support the HTML5 video element.
    </video>
  </div>

  <div class="media-box">
    <h3>2. Audio Player with Multi-Format Support</h3>
    <audio controls>
      <source src="https://interactive-examples.mdn.mozilla.net/media/cc0-audio/t-rex-roar.mp3" type="audio/mpeg">
      <source src="https://interactive-examples.mdn.mozilla.net/media/cc0-audio/t-rex-roar.ogg" type="audio/ogg">
      Your browser does not support the HTML5 audio element.
    </audio>
  </div>
</body>
</html>
```

📄 SAMPLE INPUT:

Opening multimedia.html

📄 EXPECTED OUTPUT:

Fully playable native video and audio stream with playback controls.

CSS3 Box Model, <div> Containers, Box-Shadow & Interactive Hover Transitions

🎯 AIM / OBJECTIVE:

Demonstrate the CSS3 Box Model (content, padding, border, margin), CSS display properties, box-sizing: border-box, rounded borders, and multi-layer box shadows with smooth transitions.

⚙️ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Create structured <div> container cards.
- Step 2: Explicitly style content width, padding, border, and margin.
- Step 3: Apply box-shadow with offset-x, offset-y, blur-radius, and rgba color.
- Step 4: Use :hover pseudo-class with transition: all 0.3s cubic-bezier.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 9 - CSS3 Box Model</title>
  <style>
    body { font-family: 'Segoe UI', Tahoma, sans-serif; background: #f0fdf4; padding: 40px; display: flex; justify-content: center; gap: 30px; }
    .box-card {
      width: 280px;
      padding: 24px;
      border: 2px solid #86efac;
      margin: 10px;
      border-radius: 16px;
      background: white;
      box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06);
      transition: transform 0.3s ease, box-shadow 0.3s ease, border-color 0.3s ease;
      text-align: center;
    }
    .box-card:hover {
      transform: translateY(-8px) scale(1.02);
      border-color: #22c55e;
      box-shadow: 0 20px 25px -5px rgba(34, 197, 94, 0.2);
    }
    .badge { background: #dcfce7; color: #15803d; font-weight: bold; padding: 4px 12px; border-radius: 9999px; display: inline-block; margin-bottom: 12px; }
  </style>
</head>
<body>
  <div class="box-card">
    <span class="badge">Box Model #1</span>
    <h3>Padding & Border</h3>
    <p>Content is separated from the border by 24px of internal padding space.</p>
  </div>
  <div class="box-card">
    <span class="badge">Box Model #2</span>
    <h3>Smooth Hover Effect</h3>
    <p>Hover over this card to witness elevation via cubic bezier transforms and shadows.</p>
  </div>
</body>
</html>
```

🖱️ SAMPLE INPUT:

User hovers cursor over card

🖥️ EXPECTED OUTPUT:

Smooth upward card elevation and glowing shadow expansion.

Modern CSS3 Styling (Gradients, Link Pseudo-States, Image Opacity & Borders)

🎯 AIM / OBJECTIVE:

Apply advanced CSS3 styling techniques: Linear and radial gradients, styling hypertext link states (:link, :visited, :hover, :active), image opacity filters, and custom borders.

⚙ ALGORITHM / STEP-BY-STEP PROCEDURE:

- Step 1: Define linear gradient backgrounds: linear-gradient(135deg, #6366f1, #a855f7).
- Step 2: Style link states in order (LVHA rule: Link, Visited, Hover, Active).
- Step 3: Implement image opacity transitions (opacity: 0.8 to 1.0 on hover).
- Step 4: Create custom stylish dashed/double gradient borders.

📄 SOURCE CODE IMPLEMENTATION (BCSE-013)

VERIFIED & TESTED

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>BCSE-013: Practical 10 - CSS3 Styling Masterclass</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; background: #0f172a; color: white; padding: 40px; text-align: center; }

    .hero-banner {
      background: linear-gradient(135deg, #4f46e5 0%, #7c3aed 50%, #db2777 100%);
      padding: 40px;
      border-radius: 20px;
      margin-bottom: 30px;
      box-shadow: 0 10px 30px rgba(124, 58, 237, 0.3);
    }

    a.custom-link:link { color: #38bdf8; text-decoration: none; font-weight: bold; }
    a.custom-link:visited { color: #c084fc; }
    a.custom-link:hover { color: #f43f5e; text-decoration: underline; letter-spacing: 0.5px; }
    a.custom-link:active { color: #fbbf24; }

    .styled-image {
      width: 320px;
      height: 200px;
      object-fit: cover;
      border-radius: 12px;
      border: 4px solid #38bdf8;
      opacity: 0.75;
      transition: opacity 0.4s ease, border-color 0.4s ease, transform 0.4s ease;
    }
    .styled-image:hover {
      opacity: 1.0;
      border-color: #f43f5e;
      transform: scale(1.05);
    }
  </style>
</head>
<body>
  <div class="hero-banner">
    <h1>Advanced CSS3 Styling & Visual FX</h1>
    <p>Demonstrating Multi-Stop Linear Gradients and Smooth Transitions</p>
  </div>

  <p>Learn more at: <a href="https://developer.mozilla.org" class="custom-link" target="_blank">MDN Web Docs Official Link</a></p>

  <div style="margin-top: 30px;">
    
  </div>
```

```
</body>  
</html>
```

**SAMPLE INPUT:**

User hovers link and code setup image

**EXPECTED OUTPUT:**

Dynamic color shift on link hover and crystal-clear image zoom at 100% opacity.

Project Description: Ultra-modern, highly responsive departmental web portal designed strictly using semantic HTML5 and advanced modern CSS3 (CSS Grid, Flexbox, Glassmorphism, Pure-CSS interactive tabs, accordions, and animations) aligned 100% with the official Web Technologies syllabus with zero JavaScript dependency.

Core Engineering Modules:

- ✓ 100% Pure HTML5 & Modern CSS3 (Zero JavaScript Dependency - Strictly Syllabus Aligned)
- ✓ Responsive Multi-Column CSS Grid & Fluid Flexbox Architecture
- ✓ Pure CSS Tabbed Notice Board (Radio-Driven Dynamic Circulars Switcher)
- ✓ Glassmorphic Hero Showcase & Floating KPI Statistics Badges
- ✓ Interactive Pure-CSS Accordion FAQ (<details> & <summary>) & Campus Navigator Modal (:target)
- ✓ Fully Styled Semantic HTML5 Inquiry Form with Floating Accents & Validation Constraints

 CAPSTONE ENGINE SOURCE CODE

ACADEMIC SUBMISSION READY

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Department of Computer Science & Engineering | MMEC Digital Portal</title>
  <style>
    /* =====
      DESIGN SYSTEM & CSS CUSTOM PROPERTIES (PURE CSS3)
      ===== */

    :root {
      --bg-primary: #070a12;
      --bg-secondary: #0d1322;
      --bg-card: rgba(15, 23, 42, 0.75);
      --bg-card-hover: rgba(30, 41, 59, 0.85);
      --border-subtle: rgba(255, 255, 255, 0.08);
      --border-accent: rgba(99, 102, 241, 0.45);
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
      --text-dim: #64748b;
      --primary: #4f46e5;
      --primary-light: #818cf8;
      --cyan: #06b6d4;
      --emerald: #10b981;
      --amber: #f59e0b;
      --rose: #f43f5e;
      --gradient-brand: linear-gradient(135deg, #6366f1 0%, #06b6d4 100%);
      --gradient-surface: linear-gradient(180deg, rgba(30, 41, 59, 0.7) 0%, rgba(15, 23, 42, 0.85) 100%);
      --gradient-accent: linear-gradient(135deg, rgba(99, 102, 241, 0.2) 0%, rgba(6, 182, 212, 0.1) 100%);
      --shadow-sm: 0 4px 6px -1px rgba(0, 0, 0, 0.2);
      --shadow-md: 0 10px 25px -5px rgba(0, 0, 0, 0.4);
      --shadow-glow: 0 0 30px rgba(99, 102, 241, 0.25);
      --radius-sm: 8px;
      --radius-md: 14px;
```

Q1. What is the difference between HTML and HTML5?

Answer: HTML5 is the latest major revision of HTML. Key enhancements include: semantic tags (<header>, <footer>, <nav>, <article>, <section>), native multimedia support without Flash (<audio>, <video>), Canvas and SVG graphics, client-side offline storage (localStorage, sessionStorage), and new form input types (email, date, number, range).

Q2. What is the CSS Box Model and what are its components?

Answer: The CSS Box Model is a container that wraps around every HTML element. It consists of 4 layers from inside out: 1. Content (actual text/image), 2. Padding (clears area around content, inside border), 3. Border (goes around padding), and 4. Margin (clears area outside border, creating space between elements).

Q3. What does box-sizing: border-box do?

Answer: By default (content-box), an element's width and height apply only to the content. Adding padding and border increases total rendered size. With box-sizing: border-box, the width and height properties include content, padding, and border, preventing layout breakage when adding padding.

Q4. What is the difference between localStorage, sessionStorage, and Cookies?

Answer: localStorage stores data with no expiration time across browser sessions (5-10MB). sessionStorage stores data only for the duration of the current tab/session (5MB). Cookies store small strings (up to 4KB) with expiration dates and are automatically sent to the server on every HTTP request.

Q5. Explain the order of link pseudo-classes (LVHA rule).

Answer: To function correctly without CSS cascade conflicts, link pseudo-classes must be declared in the order: :link (unvisited), :visited (visited), :hover (mouse over), and :active (clicked/pressed). Memory mnemonic: 'Love HAte'.

Q6. What is the difference between display: none and visibility: hidden?

Answer: display: none completely removes the element from the document layout flow; it takes up zero space. visibility: hidden hides the element visually, but the element still occupies its original dimensions and space in the layout.

Q7. What is the purpose of the <meta name='viewport'> tag in responsive design?

Answer: It gives the browser instructions on how to control the page's dimensions and scaling: <meta name='viewport' content='width=device-width, initial-scale=1.0'>. Without it, mobile browsers simulate desktop width (typically 980px), shrinking text and UI.

Q8. What is the difference between GET and POST HTTP request methods in forms?

Answer: GET appends form data to the URL as query parameters (?key=value); it is visible in history, bookmarked, and has length limits. POST sends data in the HTTP request body; it does not show in the URL, can send large/binary files, and is much more secure for passwords and sensitive information.

Q9. What is CSS Flexbox and how is it different from CSS Grid?

Answer: CSS Flexbox is a 1-dimensional layout system (either row OR column at a time), ideal for distributing space and aligning items in navigation bars or lists. CSS Grid is a 2-dimensional layout system (rows AND columns simultaneously), ideal for full page layouts and complex grids.

Q10. What are semantic tags and why are they important in HTML5?

Answer: Semantic tags clearly describe their meaning to both browser and developer (e.g. <header>, <nav>, <main>, <article>, <aside>, <footer>). They are crucial for SEO ranking, web accessibility (screen readers for visually impaired users), and clean maintainable code.